

V6.2 · Technical Summary and Recommendations

What was built, what it shows, and what to do next

Objective	Summarise the V6.2 semantic rebuild and set the agenda for the Ag-SERS phase.
Method	Ten analyses over the frozen atlas and the frozen V6 MSS layer, all deterministic and reproducible from results/v6_rebuild/code/.
Frozen	Atlas 09ed804a40836f4a05a91ba10900cded; the V6 MSS map M is an input, never rewritten.
Reproduce	results/v6_rebuild/code/p10 → p11 → p12 → p13

Philosophy

GAIRA is a biochemical reasoning engine, not a molecular classifier. V6.2 optimises **maximum biochemical abstraction subject to minimum information loss**. Priority order: interpretability, information retention, recoverability, and only then analyte-level accuracy — a modest accuracy loss is an acceptable price for a representation that transfers.

Headline numbers

frozen atlas	09ed804a40836f4a05a91ba10900cded — unchanged
MSS motifs	17, identical to V6
semantic levels	17 → 6 → 4
mean themes per motif	2.06 (soft, sparse)
shared motifs represented	4 of 17
multi-parent motifs in the graph	11 of 17
motif variance retained at L2	0.751
compression at L2	2.83×
MI retained at L2	1.155
motif-level recoverability	0.503
median theme confidence	0.024
Pareto-optimal K values	11 of 16

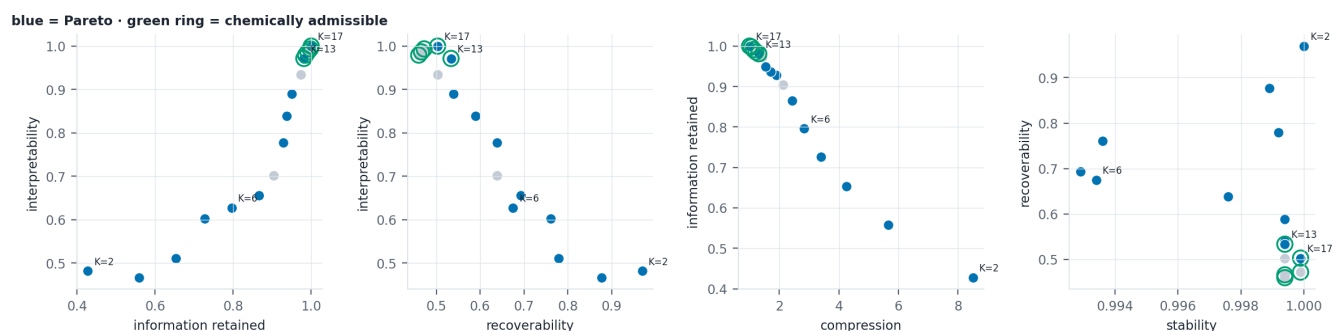


Figure 1 — The four-objective Pareto front.

What V6.2 establishes

- **Chemical overlap is real and measurable.** 15 of 17 motifs share bands with a second chemistry; 11 have more than one parent in the ontology graph. A tree was the wrong data structure.
- **Abstraction has a price curve, and it is not the accuracy curve.** Variance retained falls steeply below $K \approx 8$ while mutual information per dimension stays at or above parity down to $K=3$. Choosing K on accuracy alone would have picked a different level than choosing it on chemistry.
- **The MSS layer denoises and the theme layer does not.** Replicate variance shrinks 9× from coordinates to motifs, then grows slightly from motifs to themes.
- **Confusions are self-explaining.** Every large off-diagonal cell has a spectral cosine and a shared band list attached — except the sterol case, which is a foundation limit, not a semantic one.

What V6.2 does not establish

- That $K=6$ is optimal. It is the coarsest chemically nameable level; the Pareto front contains 11 non-dominated K values and the choice among them is a judgement.
- That the soft memberships are correct for sterol and carotenoid — both land on the wrong dominant theme, and V6.2 surfaces this rather than fixing it.
- Anything about substrates other than pure Raman. No SERS, EV, serum, bacterial or DART data enters V6.2.

Recommendations

Immediate

- Calibrate the theme posterior before any consumer treats it as a probability; the reliability curve shows it is systematically under-confident.
- Report entropy with every theme call. A soft layer whose ambiguity is discarded at the point of use is worse than a hard one.
- Add pure sterol, porphyrin and flavin references to the grounding corpus. Three of the four remaining semantic failures trace to the same absence, and no amount of work above the atlas will fix them.

For the Ag-SERS adaptation phase (not implemented here)

- **Transfer the coordinates, not the classes.** V6.2's continuous theme vector is the right target for a Raman→SERS observation model: a class label cannot be partially preserved, a coordinate can move a measurable distance.
- **Use the shared-motif list as the prior for what will be hardest.** Motifs with high membership entropy and high graph betweenness are the ones a substrate change will scramble first.
- **Keep the firewall.** Fit the observation model above the theme layer. Nothing in the atlas, M or S should be refitted on SERS data — that separation is what makes the OOD signal meaningful.
- **Gate on detectability first.** Earlier work showed 29 of 51 analytes are invisible on silver; a transfer model should be trained only on the detectable, representation-limited subset.
- **Expect the purine attractor.** On silver, 95 % of analytes previously collapsed onto purine. The V6.2 soft layer gives a natural way to express that: a substrate prior that shifts membership mass, rather than a hard re-assignment.

Artifact inventory

artifact	contents
theme_membership.yaml	soft motif→theme membership at all three levels
theme_embedding.npy	the continuous theme coordinates (167 × 6)
v62_membership.npz	S at L1/L2/L3, learned S, distances, spectra
v62_spaces.npz	embeddings, PCA, UMAP, recoverability matrix
v62_soft_hierarchy.json	levels, membership, shared motifs, uncertainty
v62_information_graph.json	bottleneck, graph, propagation, Pareto
tables/v62_*.csv	11 tables: membership, shared motifs, bottleneck, recoverability, confusion, propagation, graph, Pareto
figures_v62/*.png + *.pdf	13 figures, raster and vector
figures_v62/v62_ontology_graph.html	interactive ontology graph

Atlas 09ed804a40836f4a05a91ba10900cded verified unchanged. Reproduce: p10 → p11 → p12 → p13 under results/v6_rebuild/code/.